

# **NEBLINA 1**

## **Flight Dynamics Evidence Package**

# 1. Executive Summary & Report Status

## Evidence Package Generation Status

| Subsystem             | Status    |
|-----------------------|-----------|
| Nominal Simulation    | COMPLETE  |
| Atmosphere Model      | COMPLETE  |
| Scenario Analysis     | COMPLETE  |
| Validation Data       | COMPLETE  |
| Monte Carlo Execution | COMPLETED |

## Flight Performance

| KPI                          | Value     |
|------------------------------|-----------|
| Apogee AGL                   | 2605.1 m  |
| Max Mach                     | 0.67 M    |
| Max Velocity                 | 228.9 m/s |
| Max Total Acceleration       | 12.5 g    |
| Max Dynamic Pressure (Max-Q) | 28.3 kPa  |
| Rail Exit Velocity           | 26.4 m/s  |
| Flight Duration              | 112.7 s   |
| Ground Drift                 | 1123.5 m  |

## Stability

| KPI                          | Value    |
|------------------------------|----------|
| Static Margin (Rail Exit)    | 3.45 cal |
| Static Margin (Max-Q)        | 4.54 cal |
| Static Margin (Burnout)      | 4.66 cal |
| Max Angle of Attack (Ascent) | 60.8 deg |

## Recovery

| KPI                    | Value                                    |
|------------------------|------------------------------------------|
| Drogue Descent Rate    | 60.9 m/s                                 |
| Main Descent Rate      | 8.9 m/s                                  |
| Touchdown Velocity     | 8.8 m/s                                  |
| Touchdown Energy       | 1135.0 J                                 |
| Recovery Opening Shock | NOT PHYSICALLY RESOLVED BY CURRENT MODEL |

## Engineering Requirements Compliance

| ID               | Description                                                                            | Result    | Margin | Status           |
|------------------|----------------------------------------------------------------------------------------|-----------|--------|------------------|
| FD-LAUNCH-001    | Vehicle must leave the launch rail with sufficient velocity for aerodynamic stability. | 26.37 m/s | 1.37   | WITHIN GUIDELINE |
| FD-STABILITY-001 | Static margin must remain strictly positive during the entire burn phase.              | None      | -      | NOT EVALUATED    |
| FD-STABILITY-002 | Static margin must not exceed highly overstable limits causing extreme weathercocking. | None      | -      | NOT EVALUATED    |
| FD-LOADS-001     | Maximum airframe acceleration (thrust + aero) during ascent phase.                     | None      | -      | NOT EVALUATED    |
| REC-LOADS-001    | Maximum instantaneous opening shock deceleration.                                      | None      | -      | NOT EVALUATED    |
| REC-DESCENT-001  | Orbiter parachute terminal descent rate.                                               | None      | -      | NOT EVALUATED    |
| REC-DESCENT-002  | Main parachute terminal descent rate for safe kinetic energy ground impact.            | None      | -      | NOT EVALUATED    |
| REC-IMPACT-001   | Total kinetic energy at touchdown.                                                     | None      | -      | NOT EVALUATED    |

## 2. Model Provenance & Input Quality

### Physical Domain Validity Envelopes

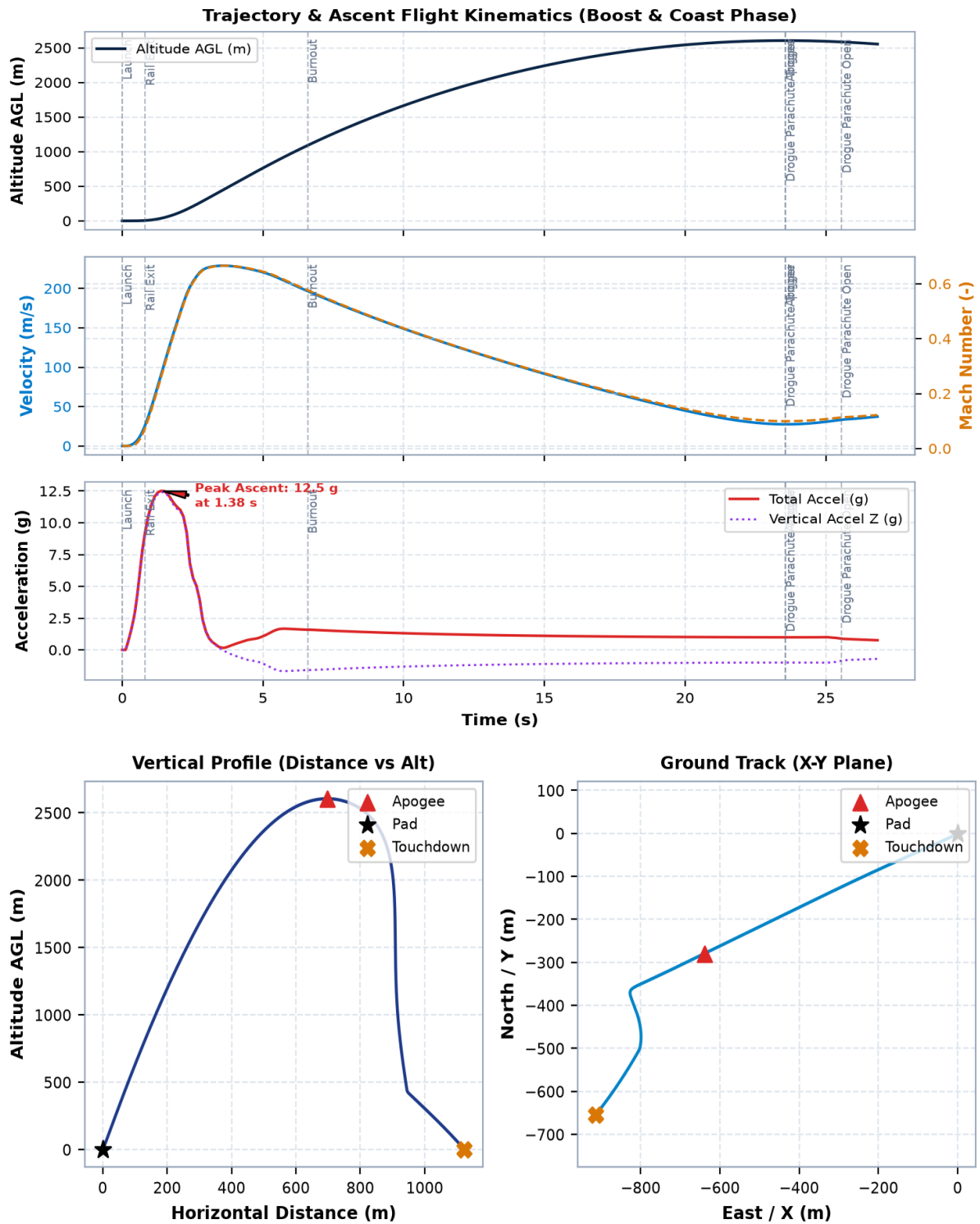
| Domain & Parameter                    | Simulated Peak | Validity Envelope               | Status            | Note                                                                                   |
|---------------------------------------|----------------|---------------------------------|-------------------|----------------------------------------------------------------------------------------|
| Aerodynamics: Flight Mach Number      | 0.67 M         | $\leq 0.80$ M (Subsonic)        | QUALIFIED         | Slender-body linear Barrowman equations valid without transonic wave drag corrections. |
| Aerodynamics: Ascent Angle of Attack  | 60.83°         | $\leq 10.0^\circ$ (Linear Lift) | NON-LINEAR REGIME | Linear normal force slope CNa assumes attached flow without crossflow separation.      |
| Atmosphere: Apogee Altitude ASL       | 3100 m         | $\leq 30,000$ m (GFS Grid)      | QUALIFIED         | MAGI weather profile covers entire trajectory domain from ground pad to apogee.        |
| Launch Dynamics: Rail Guide Clearance | 26.4 m/s       | $\geq 25.0$ m/s                 | QUALIFIED         | Rail button release dynamics assume rigid tower alignment and aerodynamic stability.   |

### Input Quality Classification

| Parameter                     | Source                             | Classification          | Confidence                     |
|-------------------------------|------------------------------------|-------------------------|--------------------------------|
| Dry Vehicle Mass              | Laboratory Precision Scale         | MEASURED                | HIGH ( $\pm 0.05$ KG)          |
| Center of Gravity (CG)        | Dual Knife-Edge Balance Rig        | MEASURED                | HIGH ( $\pm 5$ MM)             |
| Moments of Inertia (Ixx, Iyy) | SolidWorks Full Assembly CAD Model | CAD ESTIMATE            | MEDIUM ( $\pm 8\%$ )           |
| Solid Motor Thrust Curve      | Static Fire Test Bench Load-Cell   | STATIC FIRE             | HIGH ( $\pm 2\%$ IMPULSE)      |
| Propellant Initial Mass       | Pre/Post Static Fire Weighing      | MEASURED                | HIGH ( $\pm 0.02$ KG)          |
| Aerodynamic Coeffs (Cd, CNa)  | Barrowman Slender-Body Model       | OPENROCKET / ROCKETPY   | MEDIUM ( $\pm 10\%$ CD)        |
| Atmospheric Environment       | MAGI Operational GFS Forecast      | NUMERICAL WEATHER MODEL | ENSEMBLE ( $\pm 1.5$ M/S WIND) |

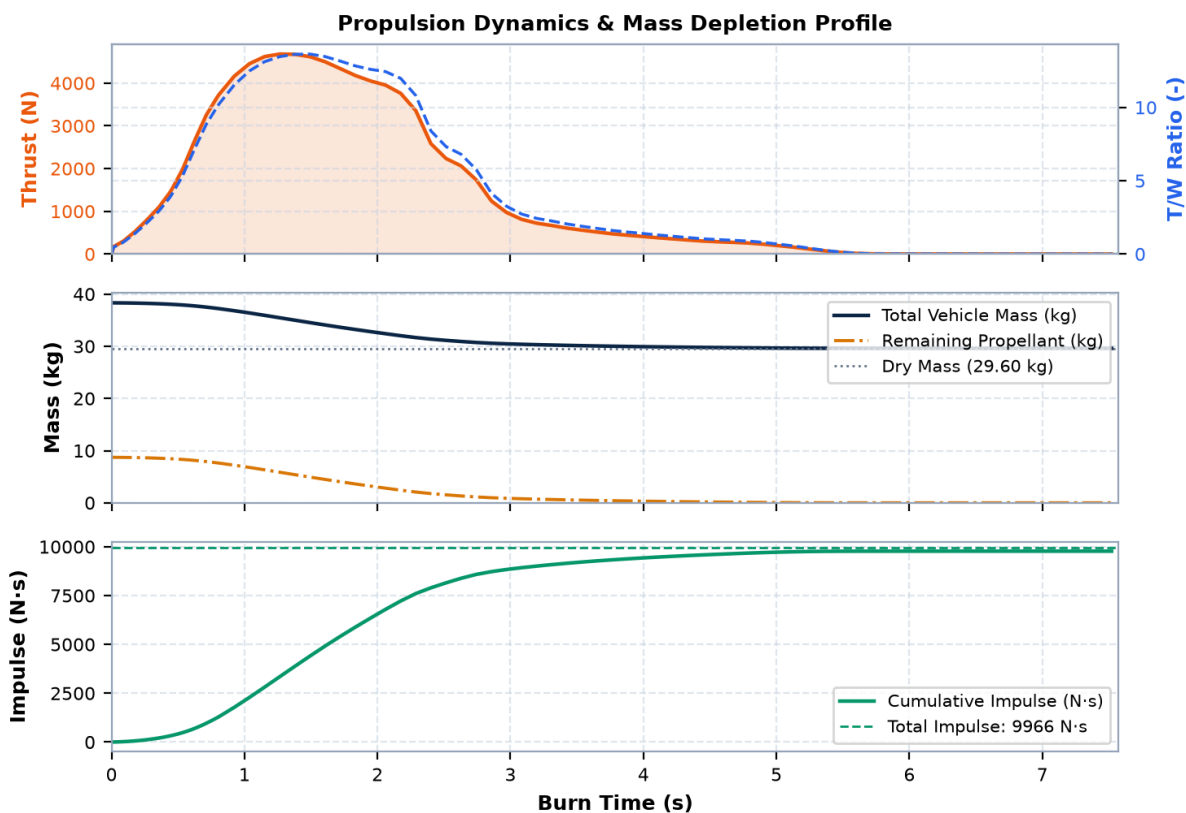
| Parameter                  | Source                                 | Classification       | Confidence           |
|----------------------------|----------------------------------------|----------------------|----------------------|
| Parachute Drag Areas (CdS) | Manufacturer Specification & Drop Test | ENGINEERING ESTIMATE | MEDIUM<br>(±12% CDS) |

### 3. Flight Kinematics & Trajectory



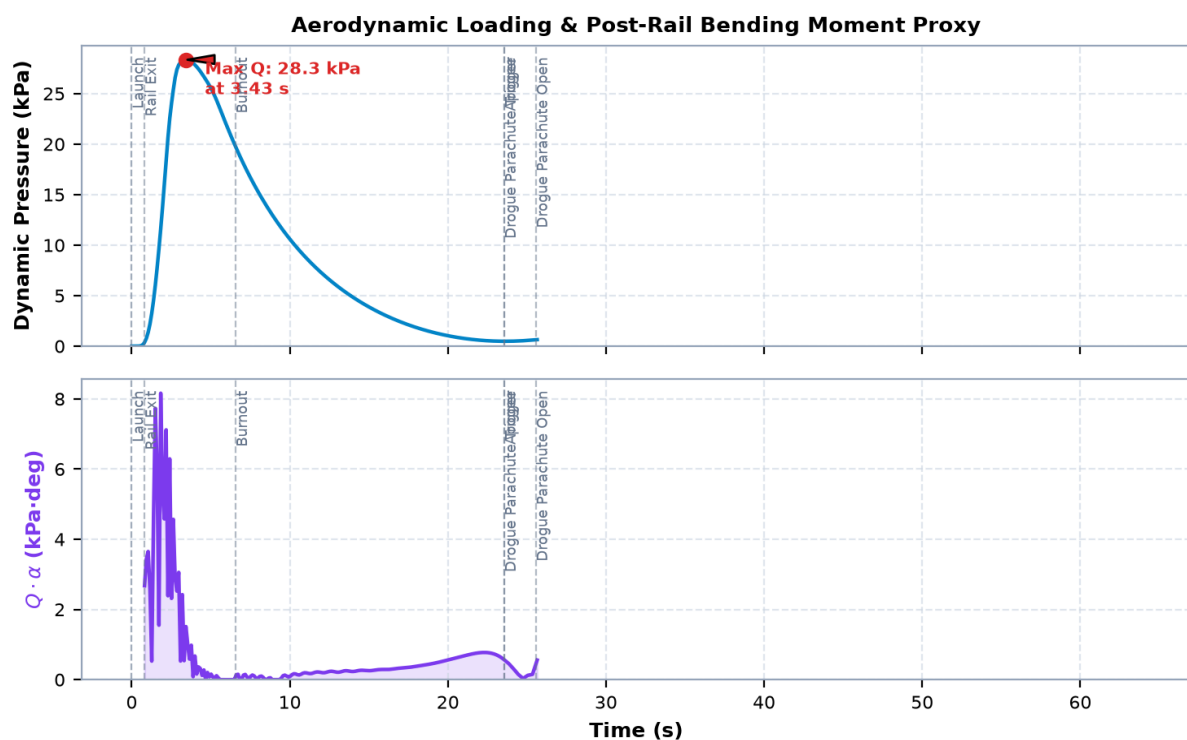
## 4. Mass & Propulsion Dynamics

| Propulsion KPI  | Value     |
|-----------------|-----------|
| Initial Mass    | 38.32 kg  |
| Burnout Mass    | 29.60 kg  |
| Propellant Mass | 8.72 kg   |
| Initial T/W     | 0.00      |
| Peak T/W        | 13.64     |
| Burn Time       | 6.59 s    |
| Total Impulse   | 9966.4 Ns |
| Average Thrust  | 1513.3 N  |
| Peak Thrust     | 4681.4 N  |



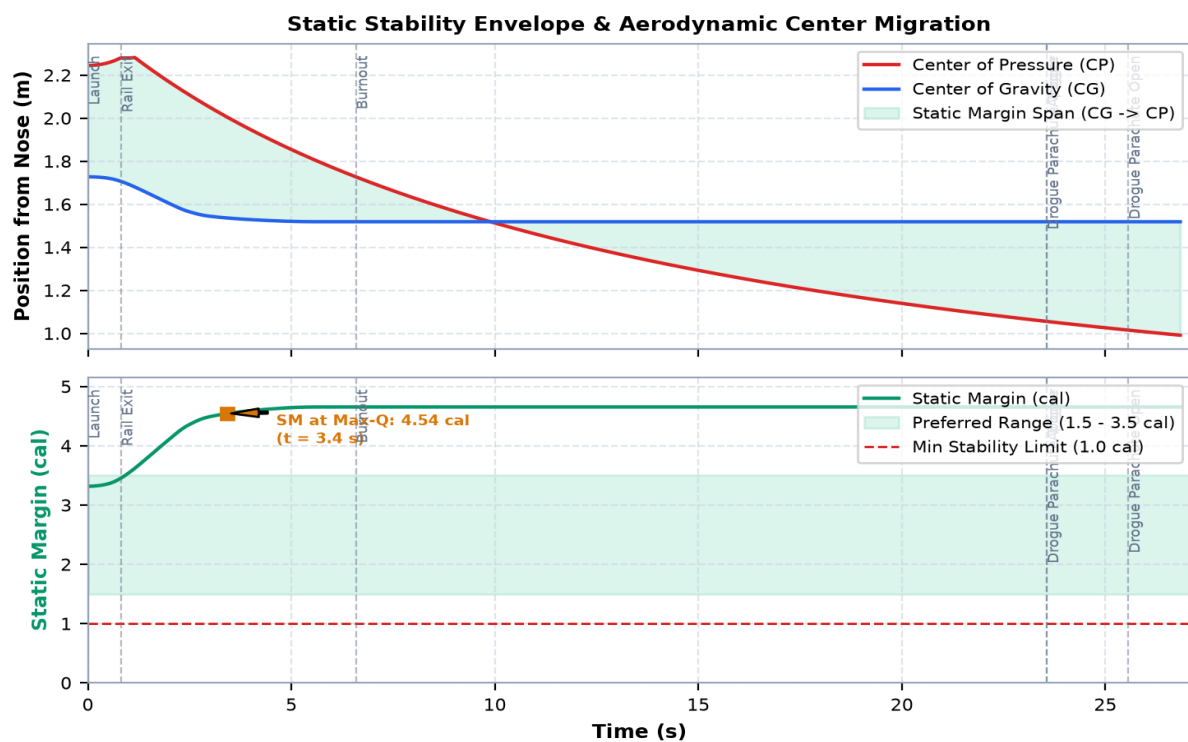
## 5. Aerodynamic Loads (Max Q & Bending)

| Max-Q State      | Value     |
|------------------|-----------|
| Dynamic Pressure | 28.3 kPa  |
| Time             | 3.43 s    |
| Altitude AGL     | 412.5 m   |
| Mach Number      | 0.67 M    |
| Velocity         | 226.1 m/s |
| Angle of Attack  | 0.1 deg   |
| Static Margin    | 4.54 cal  |



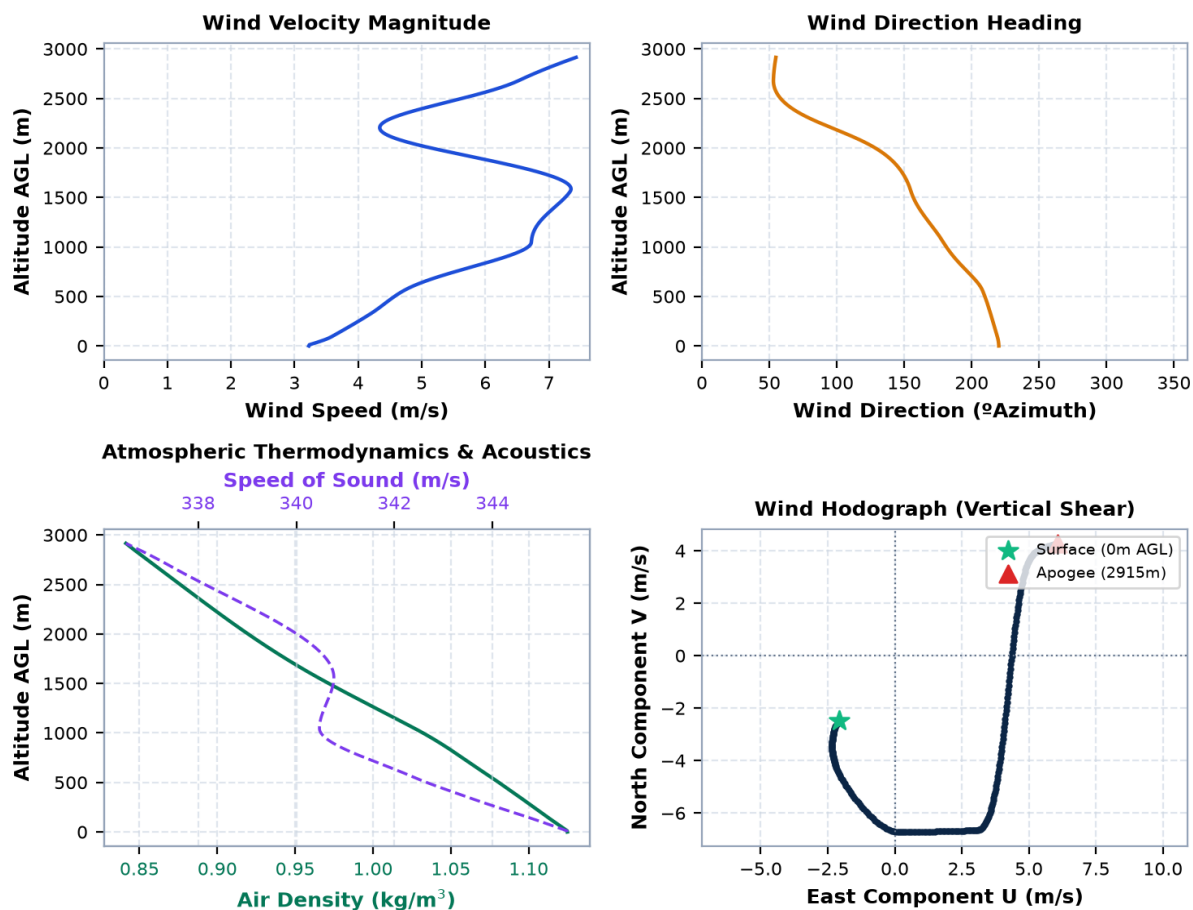
## 6. Stability Evolution

| Stability Checkpoint | Margin (cal) |
|----------------------|--------------|
| Liftoff              | 3.32         |
| Rail Exit            | 3.45         |
| Max-Q                | 4.54         |
| Burnout              | 4.66         |
| Maximum Margin       | 4.66         |

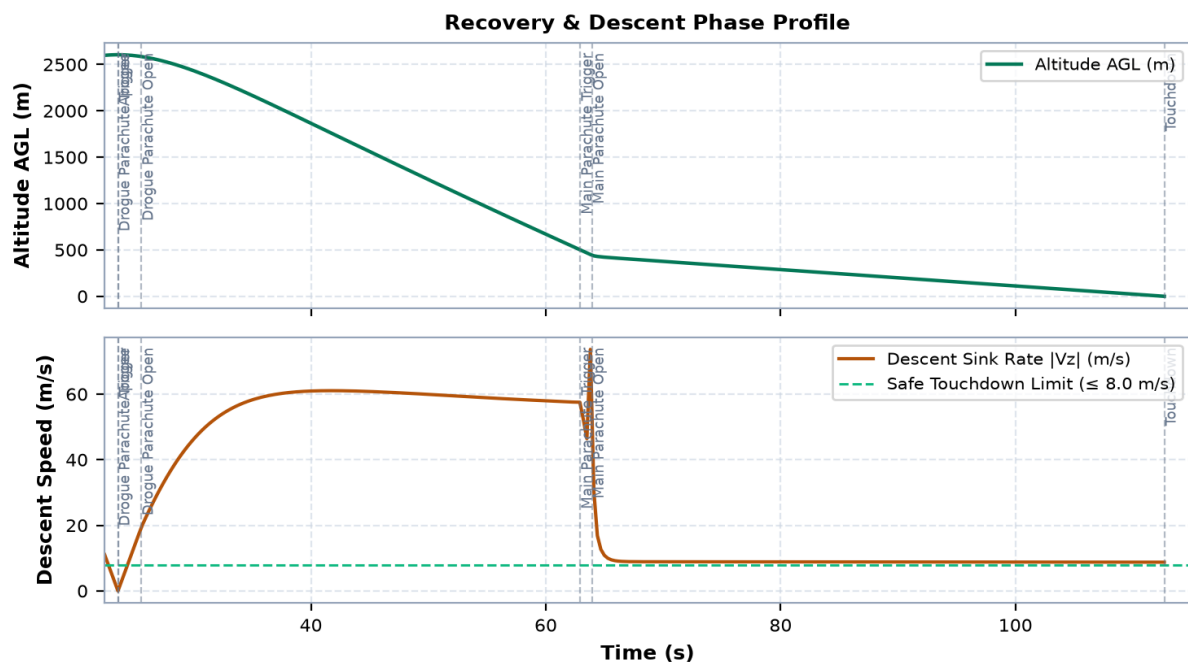


## 7. MAGI Atmospheric Profile

### MAGI High-Fidelity Atmospheric Environmental Profile

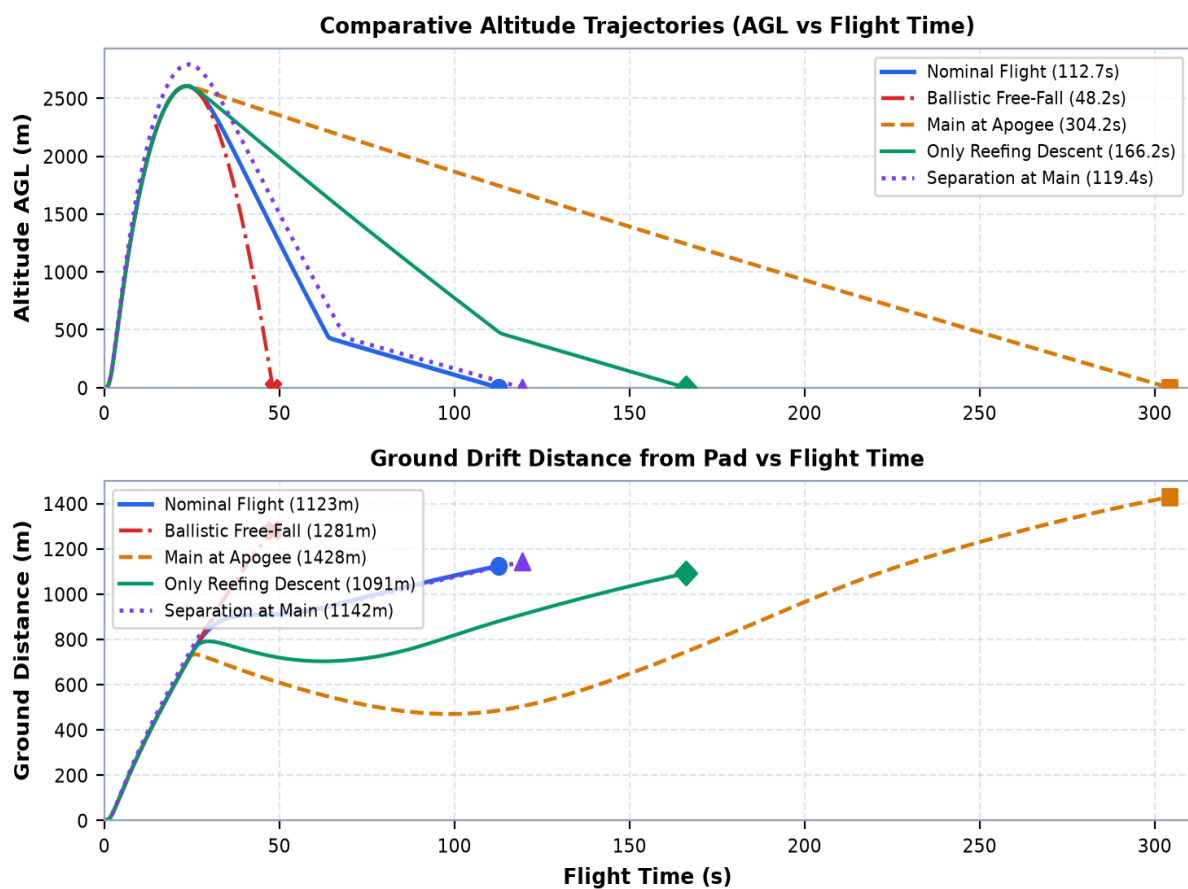


## 8. Recovery Profile



## 9. Multi-Scenario Comparative Flight

| Scenario             | Apogee (m) | Flight Time (s) | Max Mach | Max-Q (kPa) | Landing Dist (m) | Touchdown (m/s) |
|----------------------|------------|-----------------|----------|-------------|------------------|-----------------|
| Nominal Flight       | 2605.1     | 112.7           | 0.67     | 28.3        | 1123.5           | 8.8             |
| Ballistic Free-Fall  | 2605.1     | 48.2            | 0.67     | 28.3        | 1280.6           | 186.3           |
| Main at Apogee       | 2605.1     | 304.2           | 0.67     | 28.3        | 1427.9           | 8.8             |
| Only Reefing Descent | 2605.1     | 166.2           | 0.67     | 28.3        | 1091.0           | 8.8             |
| Separation at Main   | 2792.7     | 119.4           | 0.71     | 32.0        | 1142.1           | 8.5             |



## 10. Monte Carlo Stochastic Analysis

| Stochastic Run Info | Value                                 |
|---------------------|---------------------------------------|
| Campaign ID         | 2026-09-17T040233Z_neblina_1_campaign |
| Status              | COMPLETED                             |
| Requested Cases     | 20                                    |
| Completed Cases     | 20                                    |
| Failed Cases        | 0                                     |

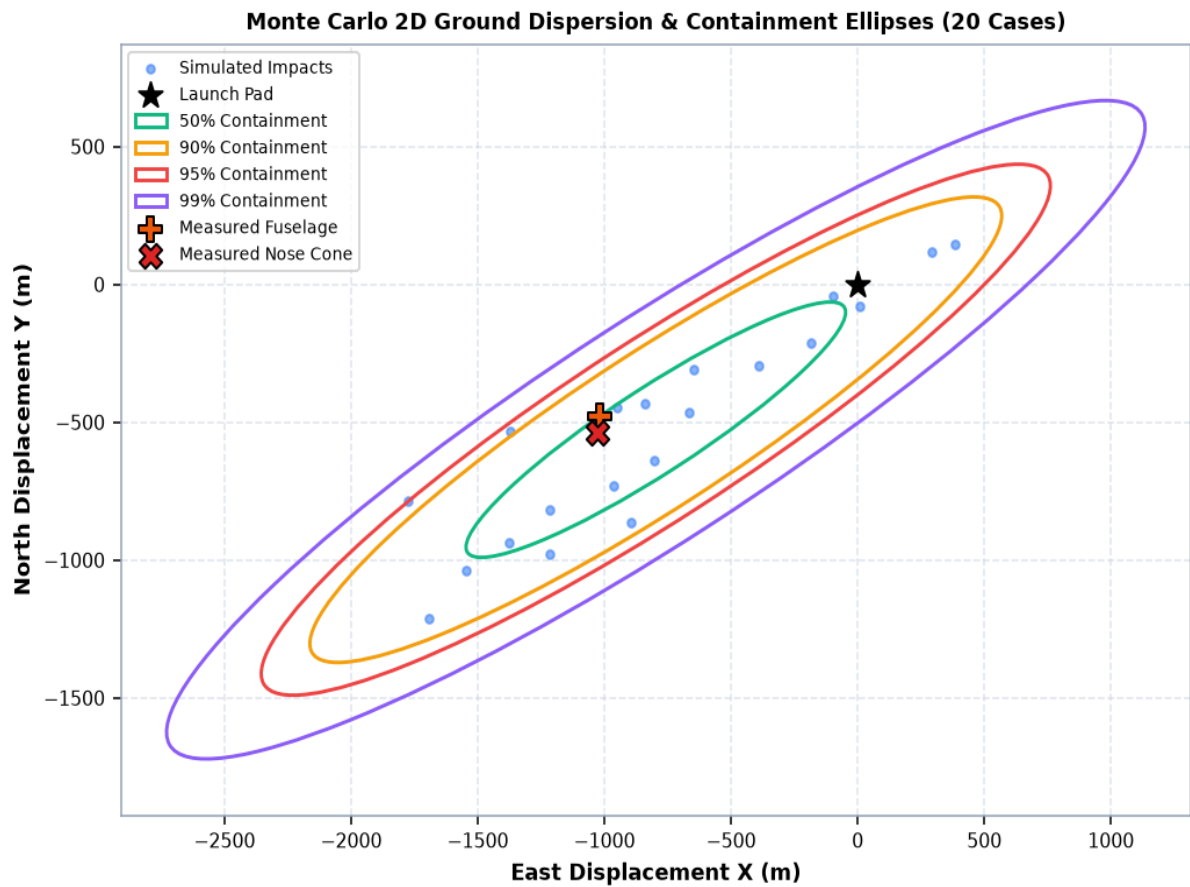
### Stochastic Inputs / Provenance

| Parameter                 | Distribution | Uncertainty Limits | Units         | Status                |
|---------------------------|--------------|--------------------|---------------|-----------------------|
| launch_inclination        | uniform      | U[70.0, 90.0]      | deg           | LEGACY_ASSUMPTION     |
| launch_heading            | uniform      | U[230.0, 250.0]    | deg           | LEGACY_ASSUMPTION     |
| power_off_drag_multiplier | normal       | N(1.0, 0.100)      | dimensionless | LEGACY_ASSUMPTION     |
| power_on_drag_multiplier  | normal       | N(1.0, 0.100)      | dimensionless | LEGACY_ASSUMPTION     |
| vehicle_mass              | normal       | N(1.0, 0.010)      | kg            | LEGACY_ASSUMPTION     |
| total_impulse_multiplier  | normal       | N(1.0, 0.050)      | dimensionless | LEGACY_ASSUMPTION     |
| atmosphere_ensemble       | categorical  | Ensemble           | categorical   | NATIVE_MODEL_ENSEMBLE |
| site_elevation            | normal       | N(1.0, 5.000)      | m             | LEGACY_ASSUMPTION     |

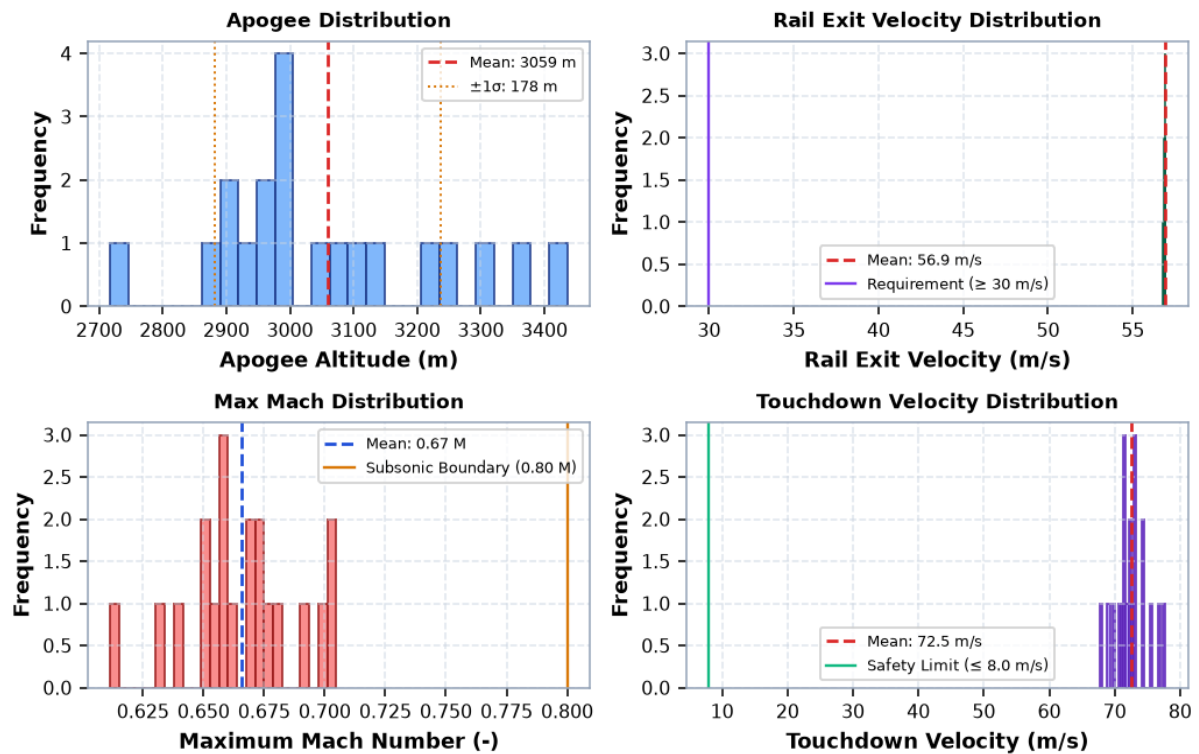
### Output Distributions

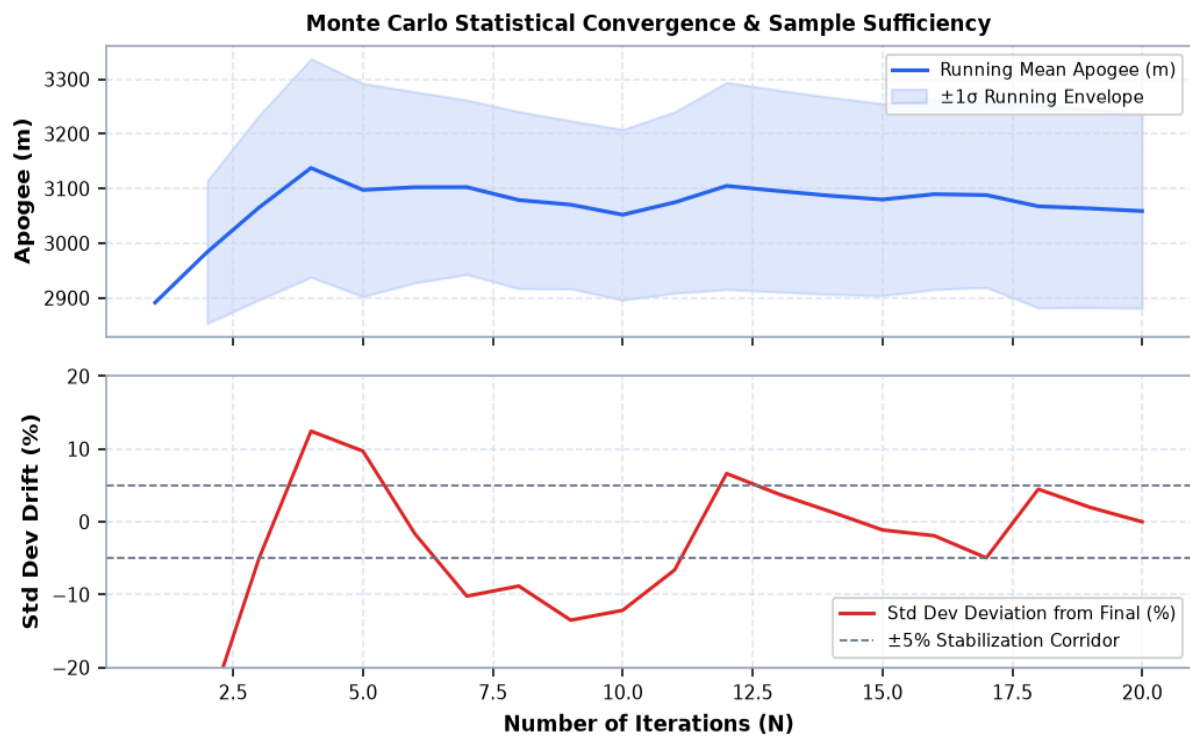
| Variable            | Mean    | Std    | P05      | P50     | P95     |
|---------------------|---------|--------|----------|---------|---------|
| Apogee AGL (m)      | 3058.94 | 173.14 | 2878.59  | 2998.46 | 3358.76 |
| Rail Exit Vel (m/s) | 56.88   | 0.05   | 56.82    | 56.89   | 56.93   |
| Max Mach (M)        | 0.67    | 0.02   | 0.63     | 0.67    | 0.70    |
| Max-Q (kPa)         | 28.37   | 1.90   | 25.66    | 28.28   | 31.43   |
| Landing X (m)       | -796.87 | 620.45 | -1695.63 | -867.30 | 295.45  |
| Landing Y (m)       | -526.94 | 383.45 | -1044.30 | -499.48 | 118.46  |
| Touchdown (m/s)     | -72.50  | 2.38   | -76.83   | -72.41  | -68.88  |
| Flight Time (s)     | 70.57   | 3.42   | 67.02    | 69.39   | 76.41   |

### Stochastic Requirement Confidence Matrix



### Monte Carlo Output Distributions & Requirement Compliance









# Reproducibility Appendix

| Attribute                 | Value                                                                      |
|---------------------------|----------------------------------------------------------------------------|
| Execution Timestamp (UTC) | 2026-09-17T04:03:06.901426Z                                                |
| Git State                 | 5bf285ed9ec5e9c9703e9f4f1ee6914f13fe0ebb (DIRTY)                           |
| Project Directory         | C:\Users\Arthur\PycharmProjects\Flight_Dynamics_Antares\projects\neblina_1 |
| Antares FD Framework      | Version 2.0 (Hardened)                                                     |

This report was automatically synthesized by Antares Flight Dynamics. No manual edits were performed on these charts or metrics. All values are traceable to the defined inputs, meteorological sources, and aerodynamic models.